

COMPARISONS OF PEAK GROUND REACTION FORCE AND RATE OF FORCE DEVELOPMENT DURING VARIATIONS OF THE POWER CLEAN

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ABSTRACT

Comfort, P, Allen, M, and Graham-Smith, P. Comparisons of peak ground reaction force and rate of force development during variations of the power clean. *J Strength Cond Res* 25(5): 1235–1239, 2011—The aim of this investigation was to determine the differences in vertical ground reaction forces and rate of force development (RFD) during variations of the power clean. Elite rugby league players ($n = 11$; age 21 ± 1.63 years; height 181.56 ± 2.61 cm; body mass 93.65 ± 6.84 kg) performed 1 set of 3 repetitions of the power clean, hang-power clean, midthigh power clean, or midthigh clean pull, using 60% of 1-repetition maximum power clean, in a randomized order, while standing on a force platform. Differences in peak vertical ground reaction forces (F_z) and instantaneous RFD between lifts were analyzed via 1-way analysis of variance and Bonferroni post hoc analysis. Statistical analysis revealed a significantly ($p < 0.001$) greater peak F_z during the midthigh power clean ($2,801.7 \pm 195.4$ N) and the midthigh clean pull ($2,880.2 \pm 236.2$ N) compared to both the power clean ($2,306.24 \pm 240.47$ N) and the hang-power clean ($2,442.9 \pm 293.2$ N). The midthigh power clean ($14,655.8 \pm 4,535.1$ N·s⁻¹) and the midthigh clean pull ($15,320.6 \pm 3,533.3$ N·s⁻¹) also demonstrated significantly ($p < 0.001$) greater instantaneous RFD when compared to both the power clean ($8,839.7 \pm 2,940.4$ N·s⁻¹) and the hang-power clean ($9,768.9 \pm 4,012.4$ N·s⁻¹). From the findings of this study, when training to maximize peak F_z and RFD the midthigh power clean and midthigh clean pull appear to be the most advantageous variations of the power clean to perform.

KEY WORDS hang power clean, mid-thigh power clean, performance

INTRODUCTION

Variations of the Clean and Power Clean are commonly incorporated into strength and conditioning programs, using different starting positions (from the floor, hang, and midthigh). The power clean has been shown to increase an athlete's performance by imitating sport-specific movements (rapid extension of the ankle, knee, and hips), while concurrently using explosive power (10). Performance in the hang-power clean has also been shown to be related to both 20-m sprint performance and countermovement jump performance (5).

Most research regarding power characteristics during performances of the clean and its variations has focused on the load that achieves the greatest peak power output (1,2,6–8). Kawamori et al. (7) determined that peak power output, during the midthigh clean pull, has been shown to be achieved at 60% of 1-repetition maximum (1RM) (power clean), although time to peak rate of force development (RFD; 99.8 ± 14.0 milliseconds) was achieved at 30% of 1RM. Time to peak RFD during midthigh clean pulls, at all loads (30, 60, 90, 120% of 1RM Power Clean), was shorter than time to peak RFD in both countermovement jump (263.3 ± 63.5 milliseconds) and vertical jump (194.7 ± 27.0 milliseconds) (7). These findings indicate that midthigh clean pulls may be preferential, over vertical jump activities, when focusing on improving RFD in an athlete.

Previously, Kawamori et al. (6) found that peak power output, during the hang-power clean, is achieved at 70% of 1RM power clean. More recently, however, Kilduff et al. (8) found that peak power output during the hang-power clean was not significantly ($p > 0.05$) different between loads of 50, 60, 70, 80, or 90% of 1RM power clean. Cormie et al. (1,2) demonstrated that peak power, during power cleans, is achieved at a load of 80% 1RM.

There is little evidence (3,4,9), however, to indicate which technique variations (power clean, hang-power clean, midthigh power clean, midthigh clean pull) may be optimal in the development of power and RFD. The evidence that is available has only determined that the second pull phase of the power clean results in the greatest vertical ground reaction forces (F_z) (3,4,9). However, no research to date has

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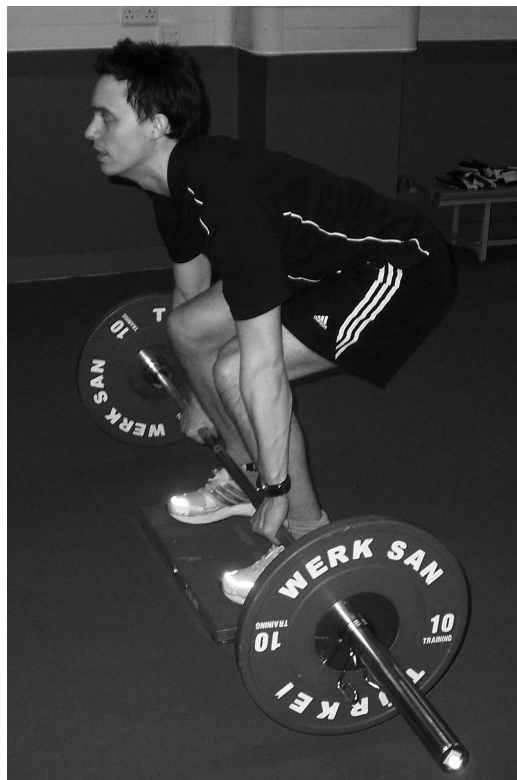


Figure 1. Power clean start position.

identified which technique variation of the power clean results in the greatest F_z or RFD.

Enoka (3) studied experienced weightlifters' techniques during the pull portion of a clean; findings showed that subjects created a peak ground reaction force (F_z) of 2,471 N during the first pull phase, whereas the second pull phase created a greater peak F_z with an average of 2,809 N. These findings were supported by Häkkinen et al. (4) who found that the second pull displayed the greatest peak F_z at 150% of the system load. More recently, Souzam et al. (10) also found that the second pull phase of the power clean resulted in significantly higher peak F_z compared to the first pull, unweighting and catch phases. However, as aforementioned, no research has compared the force characteristics of the power clean (performed from the floor—Figure 1), the hang-power clean (performed from the knee—Figure 2), the midhigh power clean (performed from midhigh—Figure 3), and the midhigh clean pull (performed from midhigh, without the catch—Figure 3). Therefore, the aim of this study was to compare peak F_z and instantaneous RFD during the power clean, hang-power clean, midhigh power clean, and midhigh clean pull. The findings of this study should aid the Strength and Conditioning Coach in determining which variation of the power clean should be used to achieve the

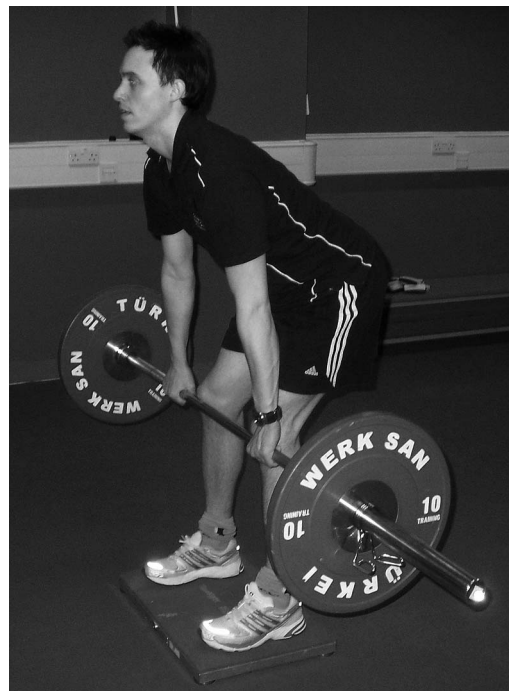


Figure 2. Hang power clean start position.

specific goals during different phases of a periodized program. It was hypothesized that the midhigh variations would result in higher peak F_z and instantaneous RFD, because previous research has identified that the second pull phase of a power clean results in the greatest F_z (3,4,9).

METHODS

Experimental Approach to the Problem

This study employed a within-subjects repeated-measures research design, whereby peak F_z and instantaneous RFD were determined during the power clean, hang-power clean, midhigh power clean, and midhigh clean pull. Peak F_z and RFD were measured by the athlete performing all lifts while standing on a force platform (Kistler, Winterthur, Switzerland, Model 9286AA, SN 1209740).

Subjects

Eleven healthy male elite rugby league players (age 20 ± 1.63 years; height 181.56 ± 2.61 cm; body mass 93.65 ± 6.84 kg) participated in this study. The investigation was approved by the Institutional Review Board, and all subjects provided informed consent before participation. The study conformed to the principles of the World Medical Association's Declaration of Helsinki. Participants had just completed preseason training and were at the end of a power mesocycle. All participants had previously conducted technique sessions to allow familiarization with the specific protocols of each lift.

Testing

Before testing all subjects performed a standardized dynamic warm-up, including each variation of the power clean (4 repetitions, 3 sets) using a standardized load (40 kg) (Ivanco Olympic bar and weights, San Pedro, CA, USA). Participants were then randomly assigned to perform 1 cluster set of 3 repetitions (30-second rest between repetitions to minimize fatigue) of each exercise, starting with the power clean, hang-power clean, midhigh power clean, or midhigh clean pull. All lifts were performed using a standardized load of 60% of previously determined 1RM power clean, because previous research has demonstrated that this load results in peak power output during midhigh cleans and hang-power cleans (6–8). All lifts were performed with subjects standing on a force plate (Kistler, Model 9286AA, SN 1209740), sampling at 1,000 Hz, interfaced with a laptop. Data were later analyzed using Bioware (Version 3.22; Kistler Instrument Corporation) to determine peak vertical ground reaction force. Instantaneous RFD was determined by dividing the difference in consecutive vertical force readings by the time interval (0.001 seconds) between readings. Data were smoothed using a moving average window of 400 milliseconds.

Statistical Analyses

Intraclass correlations were performed to determine the reliability of performances between repetitions of each exercise. A 1-way analysis of variance and Bonferroni post hoc analysis were conducted using SPSS (Version 16.0) to determine if there were any significant differences in F_z and RFD, within subjects, between lifts. The alpha level was set to $p \leq 0.05$.

RESULTS

Intraclass correlation for F_z demonstrated high reliability for each of the exercises ($r = 0.925$, $p < 0.001$, Power Clean; $r = 0.966$, $p < 0.001$, Hang-Power Clean; $r = 0.880$, $p < 0.001$, Midhigh Power Clean; $r = 0.957$, $p < 0.001$, Midhigh Clean Pull). Similarly intraclass correlations for RFD also showed high reliability for each of the exercises ($r = 0.916$, $p < 0.001$, Power Clean; $r = 0.947$, $p < 0.001$, Hang-Power Clean; $r = 0.926$, $p < 0.001$, Midhigh Power Clean; $r = 0.958$, $p < 0.001$, Midhigh Clean Pull).

One-way analysis of variance demonstrated significant differences ($p < 0.001$) in peak vertical ground reaction forces

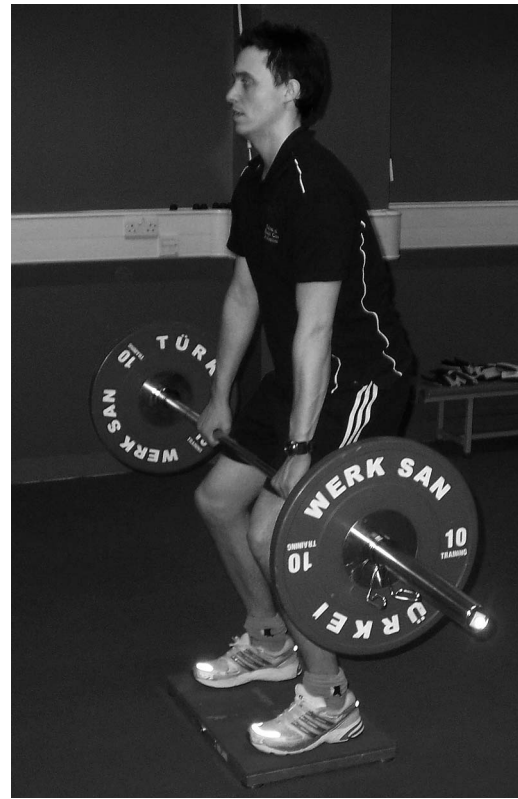


Figure 3. Mid-thigh power clean and mid-thigh pull start position.

between variations of the power clean. Bonferroni post hoc analysis revealed a significantly ($p < 0.001$) greater peak F_z during the midhigh power clean ($2,801.7 \pm 195.4$ N) and the midhigh clean pull ($2,880.2 \pm 236.2$ N) compared to both

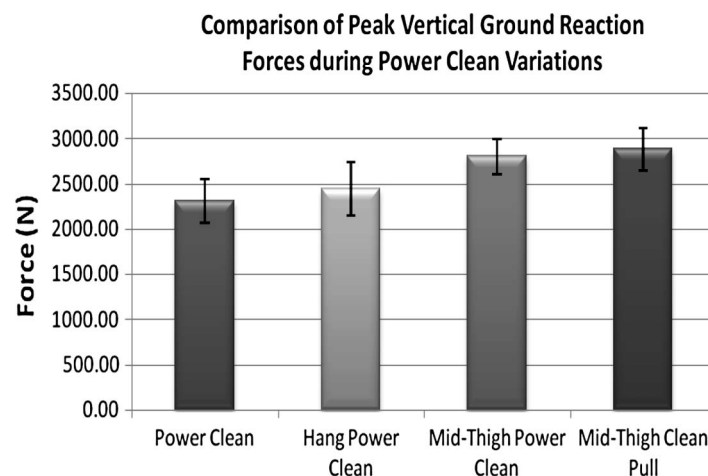


Figure 4. Comparison of peak F_z during power clean variations.

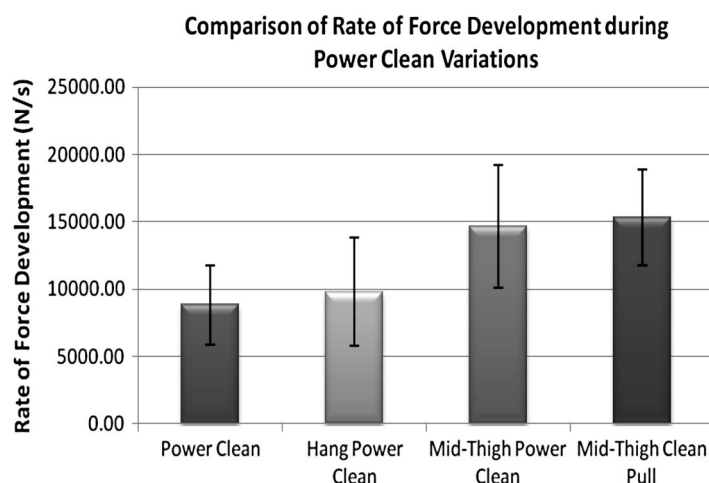


Figure 5. Comparison of peak FD during power clean variations.

the power clean ($2,306.2 \pm 240.5$ N) and the hang-power clean ($2,442.9 \pm 293.2$ N) (Figure 4).

No significant ($p > 0.05$) differences were found when comparing the peak F_z between the midthigh power clean ($2,801.7 \pm 195.4$ N) and the midthigh clean pull ($2,880.2 \pm 236.2$ N). The hang-power clean resulted in greater peak F_z ($2,442.9 \pm 293.2$ N) when compared to the power clean ($2,306.2 \pm 240.5$ N) although this was not statistically significant ($p > 0.05$) (Figure 4).

Significant differences ($p < 0.001$) in instantaneous RFD were found between variations of the power clean. Bonferroni post hoc analysis revealed a significantly ($p < 0.001$) greater RFD during the midthigh power clean ($14,655.8 \pm 4,535.1$ N·s⁻¹) and the midthigh clean pull ($15,320.6 \pm 3,533.3$ N·s⁻¹) compared to both the power clean ($8,839.7 \pm 2,940.4$ N·s⁻¹) and the hang-power clean ($9,768.9 \pm 4,012.4$ N·s⁻¹) (Figure 5).

No significant ($p > 0.05$) differences were found when comparing the RFD between either the midthigh power clean ($14,655.8 \pm 4,535.1$ N·s⁻¹) and the midthigh clean pull ($15,320.6 \pm 3,533.3$ N·s⁻¹), and between the hang-power clean ($9,768.9 \pm 4,012.4$ N·s⁻¹) compared to the power clean ($8,839.7 \pm 2,940.4$ N·s⁻¹) (Figure 4).

DISCUSSION

The results of this study demonstrate significantly ($p < 0.001$) greater peak F_z during the midthigh power clean ($2,801.7 \pm 195.4$ N) and the midthigh clean pull ($2,880.2 \pm 236.2$ N) compared to both the power clean ($2,306.2 \pm 240.5$ N) and the hang-power clean ($2,442.9 \pm 293.2$ N), when performed at a load of 60% 1RM power clean. This is in line with findings from previous research that has demonstrated that the second pull phase of the clean results in greater F_z

compared to the other phases of the lift (3,4,10). F_z during the power clean ($2,306.2 \pm 240.5$ N) and midthigh power clean ($2,801.7 \pm 195.4$ N) were also comparable to F_z during the first pull (2,471 N) and second pull (2,809 N) of the clean, respectively, in the study by Enoka (3).

Comparisons of instantaneous RFD demonstrate similar trends across the clean variations, with the midthigh power clean and the midthigh clean pull demonstrating significantly ($p < 0.001$) greater instantaneous RFD ($14,655.8 \pm 4,535.1$ and $15,320.6 \pm 3,533.3$ N·s⁻¹, respectively) than both the power clean ($8,839.7 \pm 2,940.4$ N·s⁻¹) and the hang-

power clean ($9,768.9 \pm 4,012.4$ N·s⁻¹), when performed at a load of 60% 1RM power clean.

As with the peak F_z results, there were no significant ($p > 0.05$) differences between the instantaneous RFD of the midthigh power clean ($14,655.8 \pm 4,535.1$ N·s⁻¹) and the midthigh clean pull ($15,320.6 \pm 3,533.3$ N·s⁻¹), or between the hang-power clean ($9,768.9 \pm 4,012.4$ N·s⁻¹) compared to the power clean ($8,839.7 \pm 2,940.4$ N·s⁻¹).

The similarity in peak F_z and the instantaneous RFD between the midthigh power clean and the midthigh clean pull may be explained by the fact that the concentric phases of the lifts are kinematically identical, whereas the hang-power clean and power clean exhibit noticeably different kinematics.

The midthigh power clean and the midthigh clean pull demonstrate both higher F_z and higher instantaneous RFD when compared to the hang-power clean and the power clean. It can therefore be assumed that because of a combination of greater F_z and higher instantaneous RFD that both the midthigh power clean and the midthigh clean pull result in greater peak power outputs when compared to the hang-power clean and the power clean. Furthermore, the lower limb kinematics during both of these exercises, performed from midthigh, replicate the joint angles achieved during the drive phases of both running and jumping activities. It is therefore recommended that when power output is the main goal, midthigh variations of the power clean appear to be advantageous. In contrast, the clean from the floor may be better suited to the general preparation phase of a periodized training plan.

The midthigh variations of the clean may also provide further advantages in terms of improving RFD as Kawamori et al. (7) found that time to peak RFD during midthigh clean

pulls, at all loads (30, 60, 90, 120% of 1RM Power Clean) was shorter (Peak at 30% 1RM, 99.8 ± 14.0 milliseconds) than time to peak RFD in both vertical jumps (194.7 ± 27.0 milliseconds) and countermovement jumps (263.3 ± 63.5 milliseconds). These findings and the findings of the present study indicate that midhigh clean variations may be preferential, over vertical jump activities, when focusing on improving RFD in an athlete.

Hori et al. (5) found that performance in the hang-power clean is related to both 20-m sprint performance and countermovement jump performance; however, no research has made such comparisons using the midhigh power clean and midhigh clean pull, both of which may demonstrate a stronger relationship to such performances based on the F_z and instantaneous RFD findings of this study. Furthermore, Weyand et al. (11) found that faster top running speeds are achieved through greater F_z rather than the development of limb speed; therefore, the use of midhigh power clean variations may assist with the development of sprint speed. Research into the relationship between F_z during midhigh power clean variations and top running are recommended.

It is recommended that further research determine if these findings are similar at different loads or if F_z and instantaneous RFD is greater during the hang-power clean and power clean when performed at the loads that have previously been reported to result in peak power output (70 and 80% 1RM, respectively) (1,2,6,8).

PRACTICAL APPLICATIONS

From the results of the present study, the most advantageous variations of the clean, when training to maximize F_z and RFD, appear to be the midhigh power clean and midhigh clean pull. These 2 clean variations, especially the midhigh clean pull, also offer the practical benefit that both are easier for less experienced athletes to learn and require less technical excellence. It is suggested that when optimal F_z , RFD and power output are the main goal, midhigh variations of the power clean should be used. In contrast, the clean from the

floor may be better suited to the general preparation phase of a periodized training plan, when $F_{z\max}$ RFD and peak power output are not the primary focus.

Further research investigating performance in these lifts, ideally incorporated into a training study, and their relationships to sprint and jump performance are also recommended.

REFERENCES

1. Cormie, P, McBride, JM, and McCaulley, GO. The influence of body mass on calculation of power during lower-body resistance exercises. *J Strength Cond Res* 21: 1042–1049, 2007.
2. Cormie, P, McCaulley, GO, Triplett, NT, and McBride, JM. Optimal loading for maximal power output during lower body resistance exercises. *Med Sci Sports Exerc* 39: 340–349, 2007.
3. Enoka, R. The pull in Olympic weightlifting. *Med Sci Sports* 11: 131–137, 1979.
4. Häkkinen, K, Kauhane, H, and Komi, P. Biomechanical changes in the Olympic weightlifting technique of the snatch and the clean and jerk from submaximal to maximal loads. *Scand J Sports Sci* 6: 57–66, 1984.
5. Hori, N, Newton, RU, Andrews, WA, Kawamori, N, McGuigan, MR, and Nosaka, K. Does performance of hang power clean differentiate performance of jumping, sprinting, and changing of direction? *J Strength Cond Res* 22: 412–418, 2008.
6. Kawamori, N, Crum, AJ, Blumert, PA, Kulik, JR, Childers, JT, Wood, JA, Stone, MH, and Haff, GG. Influence of different relative intensities on power output during the hang power clean: Identification of the optimal load. *J Strength Cond Res* 19: 298–308, 2005.
7. Kawamori, N, Rossi, SJ, Justice, BD, Haff, EE, Pistilli, EE, O'Bryen, HS, Stone, MH, and Haff, GG. Peak force and rate of force development during isometric and dynamic mid-thigh clean pulls performed at various intensities. *J Strength Cond Res* 20: 483–491, 2006.
8. Kilduff, LP, Bevan, H, Owen, N, Kingsley, MIC, Bunce, P, Bennett, M, and Cunningham, D. Optimal loading for peak power output during the hang power clean in professional rugby players. *Int J Sports Physiol Perf* 2: 260–269, 2007.
9. Souzain, AL, Shimada, SD, and Koontz, A. Ground reaction forces during the power clean. *J Strength Cond Res* 16: 423–427, 2002.
10. Stone, MH. Explosive exercise and training. *Natl Strength Cond Assoc J* 15: 7–15, 1993.
11. Weyand, PG, Sternlight, DB, Bellizzi, MJ, and Wright, S. Faster top running speeds are achieved with greater ground forces not more rapid leg movements. *J Appl Physiol* 89: 1991–1999, 2000.